Ofqual QAN	**600/3055/0**
Quartz ID	**1158, 1159, 1161**
Qualification	**Level 5 Diploma in Computing, Level 5 Diploma in Computing with Business Management, Level 5 Diploma in Computing with Data Science**

Assessment	**Introduction to Data Science and Big Data**
Assessment Type	**Examination**
Date	**Wednesday 28th May 2025**
Time	**9.00am – 12.00pm**

Marking Scheme

Markers are advised that, unless a task specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) must be given the available marks. If there is doubt as to the correctness of an answer, the relevant NCC education materials should be the first authority.

This marking scheme has been prepared as a guide only to markers and there will frequently be many alternative responses which will provide a valid answer.

Each candidate's script must be fully annotated with the marker's comments (where applicable) and the marks allocated for each part of the tasks.

Throughout the marking, please credit any valid alternative point.

Marker's comments:		
Moderator's comments:		
Mark:	**Moderated mark:**	**Final mark:**
Penalties applied for academic malpractice:		

Answer ALL questions

Marks

Question 1

a) Discuss the differences between a Data Scientist, a Data Analyst and a Big Data Expert, focusing on their roles **and** responsibilities. **6**

Marking scheme
Data Scientist: Role: Focuses on building predictive models, conducting statistical analyses, and utilising machine learning to extract insights from data. (1 Mark) Responsibility: Involves designing algorithms, working with unstructured and structured data, and presenting actionable insights. (1 Mark)

Data Analyst: Role: Primarily responsible for data interpretation and visualisation to help stakeholders make data-driven decisions. (1 Mark) Responsibility: Cleaning and analysing data, identifying trends and patterns, and preparing reports or dashboards. (1 Mark)

Big Data Expert: Role: Focuses on managing and processing massive datasets using distributed systems and big data technologies. (1 Mark) Responsibility: Designs data pipelines, works with tools like Hadoop and Spark, and ensures scalability and efficiency in data handling. (1 Mark)). Any suitable alternative allowed, 1 mark for each correct point to a maximum of 6 marks.

b) Select TWO (2) of the roles in part (a). For each role highlight unique key skills and explain how these skills contribute to their work. **4**

Marking scheme
Data Scientist: Skill: Proficiency in machine learning and statistical modelling. Contribution: Enables the creation of predictive models and algorithms for extracting insights. (1 Mark)
Skill: Expertise in programming languages like Python or R.
Contribution: Facilitates data analysis, cleaning, and visualisation effectively. (1 Mark)
Data Analyst: Skills: Strong data visualisation skills. Contribution: Helps in creating clear, interactive dashboards and reports for stakeholders. (1 Mark)
Skills: Proficiency in querying databases using SQL.
Contribution: Allows efficient extraction and manipulation of data for analysis. (1 Mark)
Big Data Expert: Skills: Knowledge of distributed computing systems like Hadoop or Spark.
Contribution: Enables the processing and management of large datasets effectively. (1 Mark)
Skills: Understanding of NoSQL databases like Cassandra or MongoDB.
Contribution: Facilitates the handling of unstructured or semi-structured data for scalability. (1 Mark)
Any suitable alternative allowed, 1 mark for correct each skill and contribution to a maximum of 4 marks.

Total 10 Marks

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Marks

Question 2

a) Identify **and** describe **THREE (3)** main sources of data in the modern digital world **and** provide **ONE (1)** example for each. **6**

Marking scheme

Social Media Platforms: Platforms like Facebook, Twitter, and Instagram generate massive amounts of user-generated content, including posts, comments, and reactions. (1 Mark) Examples: Tweets analysed for sentiment analysis. (1 Mark)

IoT Devices: Devices such as smart sensors, wearables, and smart home devices collect real-time data on user behaviour and environmental conditions. (1 Mark) Example: Data from fitness trackers like Fitbit used to monitor health trends. (1 Mark)

Web Logs and Clickstream Data: Data collected from website interactions, such as page views, clicks, and navigation patterns. (1 Mark) Example: Data used by Amazon to recommend products. (1 Mark)

Business Transactions: Point-of-sale systems, e-commerce platforms, and payment gateways record transactional data such as purchases and payments. (1 Mark) Example: Sales data used for inventory management in retail. (1 Mark)

Any suitable alternative allowed, 1 mark for data source and 1 mark for corresponding example to a maximum of 6 marks.

b) Discuss **TWO (2)** challenges associated with data quality **and** their potential solutions. **4**

Marking scheme

- ***One major challenge associated with data quality is missing or incomplete data, which occurs when certain values are absent from a dataset, making analysis unreliable (1 Mark). This can be addressed through data imputation techniques, such as replacing missing values with the mean, median, or mode, or using predictive models to estimate missing data points (1 Mark).***
- ***Another common challenge is data inconsistency, where different sources provide conflicting information, leading to discrepancies in analysis (1 Mark). To resolve this, organisations implement data standardisation and validation rules that enforce uniform formatting and consistency across datasets, ensuring that data remains accurate and reliable for decision-making (1 Mark).***

Any suitable alternative allowed such as Data duplication, data outlier etc, 1 mark for challenges and 1 mark for solution per bullet point to a maximum of 4 marks.

Total 10 Marks

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Marks

Question 3

a) List **and** define the **SIX (6) broad categories** of data types used in data science. **6**

Marking scheme

Numerical Data – Example: A dataset containing the heights of students in centimetres (e.g., 170 cm, 165 cm) (1 Mark).

Categorical Data – Example: Different types of fruits such as Apple, Banana, and Orange (1 Mark).

Textual Data – Example: Customer reviews on an e-commerce website like "This product is excellent!" (1 Mark).

Temporal Data – Example: Daily temperature readings recorded at 12 PM for a month (1 Mark).

Multimedia Data – Example: A video file of a lecture recording or an image uploaded on Instagram (1 Mark).

Derived Data – Example: Average sales revenue computed from daily sales figures over a month (1 Mark).

Any suitable alternative allowed, 1 mark for each pair of a correct data type and its corresponding example to a maximum of 6 marks.

b) Explain how understanding data types is essential in data preprocessing and analysis. **4**

Marking scheme

Understanding data types is crucial in data preprocessing and analysis because it determines how data should be processed, cleaned, and analysed (1 Mark). Different data types require specific preprocessing techniques (1 Mark); for instance, numerical data may need normalisation or scaling (1 Mark), while categorical data requires encoding methods like one-hot encoding (1 Mark). Additionally, selecting appropriate statistical and analytical techniques depends on the data type—textual data requires Natural Language Processing (NLP) methods (1 Mark), while temporal data needs time-series analysis techniques (1 Mark). Proper identification of data types ensures accurate data visualisation (1 Mark), prevents errors in analysis, and enhances model performance by applying the right transformations (1 Mark).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 4 marks.

Total 10 Marks

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Marks

Question 4

a) Define Central Tendency **and** explain its role in Exploratory Data Analysis (EDA). **4**

Marking scheme
Central Tendency refers to the statistical measure that gives a "centre or single value" around which the measurements in the data are distributed (1 Mark). In EDA, central tendency plays a key role in understanding the dataset's distribution (1 Mark), detecting skewness (1 Mark), and identifying anomalies or outliers (1 Mark). By summarising data into a central value, it allows analysts to make initial observations (1 Mark) and compare different datasets effectively (1 Mark), helping in decision-making and model selection (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 4 marks.

b) Provide examples of **THREE (3) measures** of central tendency, explain their definitions **and** describe their applications. **6**

Marking scheme

- ***Mean (Arithmetic Average) – The sum of all values divided by the number of observations (1 Mark). Used in salary analysis to determine the average income of employees in a company (1 Mark).***
- ***Median (Middle Value) – The middle value when data is arranged in ascending or descending order (1 Mark). Used in real estate pricing to find the typical house price, reducing the impact of extreme values (1 Mark).***
- ***Mode (Most Frequent Value) – The value that appears most frequently in a dataset (1 Mark). Used in market research to identify the most preferred product size or colour among consumers (1 Mark).***

Any suitable alternative allowed, 2 marks per bullet point: 1 mark for definition and 1 mark for application, to a maximum of 6 marks.

Total 10 Marks

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Marks

Question 5

a) What is binning in the context of data preprocessing? **7**
Explain the difference between THREE (3) different binning techniques with an example of each.

Marking scheme
Binning is a data preprocessing technique used to group continuous or categorical data into discrete intervals, making analysis easier and reducing noise (1 Mark).
Categorical Binning: Groups categorical data into broader categories to simplify analysis (1 Mark). Grouping customer age ranges as teen (13-19), Adult (20-59), Senior (60+) instead of individual ages. (1 Mark)
Numerical Binning: Converts continuous numerical data into fixed intervals or bins (1 Mark). Converting exam scores (0-100) into bins like Fail (0-40), Pass (41-70), Excellent (71-100). (1 Mark)
Quantile Binning: Divides numerical data into equal-sized bins based on percentiles (1 Mark). Income data divided into quartiles: Low (0-25%), Medium (26-50%), High (51-75%), Very High (76-100%). (1 Mark)

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 7 marks.

b) Define class imbalance in datasets. **3**
Why is it a challenge for machine learning models?

Marking scheme
Class imbalance occurs when one class in a dataset has significantly more samples than another, leading to skewed distributions (1 Mark).
Class imbalance is a challenge because machine learning models tend to be biased (1 Mark) towards the majority class, resulting in poor predictive performance for the minority class (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 3 marks.

Total 10 Marks

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Marks

Question 6

a) Explain the concept of Reinforcement Learning (RL) **and** discuss how it differs from Supervised Learning. **5**

Marking scheme
RL is a machine learning approach where an agent interacts with an environment and learns by receiving rewards or penalties for its actions (1 Mark). The goal is to maximise cumulative rewards through trial and error (1 Mark), making RL useful in robotics, gaming, and autonomous systems (1 Mark).

Unlike Supervised Learning, which learns from labelled data with clear input-output pairs (e.g., email spam detection) (1 Mark), RL does not rely on predefined labels (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks.

b) Provide **ONE (1)** real-world example where Reinforcement Learning is applied, explaining how it improves decision-making. **5**

Marking scheme
One real-world application of Reinforcement Learning (RL) is in self-driving cars (1 Mark), where the system learns to navigate roads safely by interacting with its environment (1 Mark). The car's AI agent continuously makes decisions, such as adjusting speed, changing lanes, or braking, based on inputs from sensors and cameras (1 Mark). The RL model is trained using a reward-based system—safe driving actions, like maintaining a safe distance from other vehicles, are rewarded, while dangerous actions, such as sudden braking or swerving, incur penalties (1 Mark). Over time, the system refines its decision-making, improving driving efficiency and safety (1 Mark). By learning from real-time experiences, RL enables self-driving cars to adapt to different traffic conditions, reduce accidents, and optimise fuel consumption, making autonomous driving more reliable and effective (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks.

Total 10 Marks

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Marks

Question 7

a) Describe the **THREE (3)** phases of understanding in data visualisation: perceiving, interpreting and comprehending. **6**

Marking scheme

- ***Perceiving, involves visually identifying the elements of a chart and answering the question, "What do I see?" (1 Mark). This includes recognising the type of chart being used, understanding what data the marks represent, and identifying the range of values displayed (1 Mark). Additionally, it involves assessing value associations and determining whether the data representation is trustworthy (1 Mark).***
- ***Interpreting, translates visual observations into quantitative or qualitative meaning, answering, "What does this mean?"(1 Mark). It requires analysing features such as patterns, trends, and differences, as well as identifying expected or unexpected results (1 Mark). It also involves determining which features are most important given the subject of the chart (1 Mark).***
- ***Comprehending focuses on achieving a deeper understanding by synthesising insights from the data to draw conclusions (1 Mark). This phase answers the question, "What conclusions can I draw?" and includes validating the interpreted information (1 Mark), connecting it to actionable decisions, and using the findings to address the problem or decision context (1 Mark).***

Any suitable alternative allowed, 2 marks per bullet to a maximum of 6 marks.

b) Explain how these phases contribute to deriving meaningful insights from data. **4**

Marking scheme

The three phases of understanding in data visualisation—Perceiving, Interpreting, and Comprehending—work together to transform raw data into actionable insights (1 Mark). The Perceiving phase helps identify the structure and key elements of the data, such as patterns, trends, or anomalies (1 Mark), which lays the foundation for deeper analysis (1 Mark). The Interpreting phase builds on this by adding meaning to these observations (1 Mark), helping analysts connect the visual features to their context or subject matter, such as determining whether a trend is expected or significant (1 Mark). Comprehending phase allows decision-makers to synthesise these insights, validate their relevance, and apply them to real-world problems or strategies (1 Mark), ensuring that the data leads to informed decisions and actionable outcomes (1 Mark).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 4 marks.

Total 10 Marks

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Marks

Question 8

a) Define Mean Absolute Error (MAE) **and** explain its importance in regression analysis. **4**

Marking scheme
MAE is a metric used to measure the average magnitude of errors between predicted and actual values in regression models (1 Mark). It is calculated as the mean of the absolute differences between the predicted and observed values (1 Mark), ensuring that all errors are treated equally regardless of direction (1 Mark). MAE is important in regression analysis because it provides a straightforward interpretation of the model's prediction accuracy, expressed in the same units as the target variable, making it intuitive and easy to understand (1 Mark). MAE helps analysts assess how well a model performs and identify areas for improvement (1 Mark). Additionally, its sensitivity to large errors is lower than other metrics like Mean Squared Error (MSE), making it a preferred choice when outliers are less significant in decision-making (1 Mark).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 4 marks.

b) Define AUC (Area Under the ROC Curve) **and** explain its importance in classification problem evaluation. **6**

Marking scheme
AUC is a performance metric used in classification problems to evaluate the ability of a model to distinguish between positive and negative classes (1 Mark).
It measures the area under the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 Mark).
The AUC value ranges from 0 to 1 (1 Mark), where a higher AUC indicates better model performance (1 Mark), with 1 representing perfect classification and 0.5 signifying no better than random guessing (1 Mark). AUC is important because it provides a comprehensive evaluation of a model's performance across various classification thresholds rather than focusing on a single point, making it particularly useful in imbalanced datasets (1 Mark). By measuring how well a model separates the positive and negative classes, it ensures that the evaluation is threshold-independent (1 Mark) and highlights the trade-off between sensitivity and specificity (1 Mark). This makes AUC a key metric for comparing different models (1 Mark) and selecting the one that performs best in distinguishing classes across all possible decision thresholds (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 6 marks.

Total 10 Marks

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Marks

Question 9

a) Discuss **TWO (2)** components of Business Intelligence architecture: Data Warehousing and Business Analytics. Use examples to illustrate your answer. **5**

Marking scheme
Data Warehousing is a centralised repository that stores data from multiple sources (1 Mark), both historical and real-time, to support analysis and reporting (1 Mark).
For example, a retail company uses data warehousing to store sales data for performance tracking (1 Mark). Business Analytics involves analysing this data through methods like querying, reporting, and predictive modelling to extract actionable insights (1 Mark). For instance, analytics can help forecast customer demand based on historical purchasing trends (1 Mark). Together, these components enable data-driven decision-making (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks.

b) Explain how Data Warehousing **and** Business Analytics contribute to informed decision-making with examples. **5**

Marking scheme
Data Warehousing contributes to informed decision-making by providing a centralised and organised repository for historical and real-time data (1 Mark), enabling businesses to analyse trends and patterns effectively (1 Mark). For example, a retail company can use a data warehouse to identify regional sales trends, optimise inventory, and improve supply chain efficiency (1 Mark).
Business Analytics enhances decision-making by applying statistical and computational methods to extract actionable insights from data (1 Mark). For instance, predictive analytics can help a company forecast customer demand (1 Mark), while reporting tools enable management to monitor key performance indicators in real-time (1 Mark). Together, these components ensure decisions are based on accurate, up-to-date, and comprehensive data (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks.

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Marks

Question 10

a) Explain the concepts of Privacy, Fairness and Transparency in the context of Data Ethics. Use examples to illustrate your answer. **5**

Marking scheme
Privacy refers to the right of individuals to control how their personal data is collected, stored, and shared, ensuring it is used only for authorised purposes (1 Mark). For example, protecting user data in online transactions is crucial to prevent misuse (1 Mark).
Fairness emphasises that data processes and decisions must not discriminate or introduce bias against individuals or groups (1 Mark). For instance, AI hiring tools must ensure decisions are unbiased and equitable across all demographics (1 Mark).
Transparency involves making data practices, algorithms and decision-making processes clear and understandable to stakeholders (1 Mark). For example, providing clear explanations of how a credit scoring system works builds trust and accountability (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

b) Explain why maintaining fairness and transparency is critical in data-driven decision-making. **5**
Provide an example where a lack of these aspects leads to ethical concerns.

Marking scheme
Maintaining fairness in data-driven decision-making is essential to prevent biased outcomes that could disadvantage certain individuals or groups (1 Mark). Bias in algorithms can lead to discriminatory hiring practices, unfair loan approvals, or unequal access to services (1 Mark). Transparency ensures that decision-making processes are understandable, accountable, and free from hidden biases (1 Mark), increasing public trust in AI and data systems (1 Mark).
A notable example of ethical concerns due to a lack of fairness and transparency is AI hiring tool, which was found to favour male candidates over female applicants due to biased training data (1 Mark).
Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

Total 10 Marks

End of paper

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Learning Outcomes matrix

Questions	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO 1	Yes
2	LO 1, 2	Yes
3	LO 2, 3	Yes
4	LO 2, 3	Yes
5	LO 2, 3, 4	Yes
6	LO 1, 3, 4	Yes
7	LO 2, 4, 5	Yes
8	LO 4, 5	Yes
9	LO 1,4, 5	Yes
10	LO 1, 4, 6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Demonstrate a systematic and thorough understanding of the concept of Data Science and Big Data	Exhibits minimal grasp of foundational concepts in Data Science and Big Data. Fails to showcase an ability to identify, adapt, or utilize appropriate skills, methods, and procedures effectively in addressing data-related challenges.	Displays an insufficient understanding of essential concepts in Data Science and Big Data. Struggles to articulate and apply appropriate skills, methods, and procedures adequately, indicating a need for further development.	Achieves a satisfactory level of understanding in the fundamental concepts of Data Science and Big Data. Demonstrates the ability to identify, adapt, and apply foundational skills, methods, and procedures to address standard data challenges.	Attains a very good understanding of advanced concepts in Data Science and Big Data. Exhibits a proficient ability to identify, adapt, and effectively use a range of skills, methods, and procedures, surpassing basic requirements.	Attains an outstanding level of comprehension in intricate concepts of Data Science and Big Data. Demonstrates an exceptional ability to identify, adapt, and apply a comprehensive set of skills, methods, and procedures, yielding well-supported, highly nuanced solutions that showcase depth and mastery.
Demonstrate proficiency in data collection, design, and modelling for data processing	Struggles to demonstrate basic proficiency in data collection, design, and modelling for data processing. Exhibits limited understanding of fundamental concepts and is unable to apply them effectively.	Displays some proficiency in data collection, design, and modelling for data processing but falls short of the expected standard. Shows inconsistencies or weaknesses in applying relevant skills, methods, and procedures.	Demonstrates a satisfactory level of proficiency in data collection, design, and modelling for data processing. Applies foundational skills, methods, and procedures effectively, achieving competence in standard scenarios.	Attains a very good level of proficiency in data collection, design, and modelling for data processing. Demonstrates a well-developed ability to apply a range of skills, methods, and procedures effectively, surpassing basic requirements.	Achieves an exceptional level of proficiency in data collection, design, and modelling for data processing. Displays mastery in applying a comprehensive set of advanced skills, methods, and procedures, showcasing depth and excellence in addressing complex data processing challenges.

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Select appropriate techniques and tools for data pre-processing.	Struggles to select or apply basic techniques and tools for data pre-processing	Shows some ability to select techniques and tools but falls short of the expected standard.	Demonstrates a satisfactory ability to select appropriate techniques and tools for data pre-processing.	Attains a very good ability to select and apply a range of techniques and tools for data pre-processing.	Achieves an exceptional ability to select and apply advanced techniques and tools for data pre-processing.
Understand the analytical techniques and software tools to effectively generate insights from data.	Struggles to select or apply basic techniques and tools for data pre-processing	Shows some ability to select techniques and tools but falls short of the expected standard.	Demonstrates a satisfactory ability to select appropriate techniques and tools for data pre-processing.	Attains a very good ability to select and apply a range of techniques and tools for data pre-processing.	Achieves an exceptional ability to select and apply advanced techniques and tools for data pre-processing.
Understand the suitability of different visualisation methods for different types of Data	Struggles to understand the basic suitability of visualization methods for different types of data.	Displays some understanding of the suitability of visualization methods but falls short of the expected standard.	Demonstrates a satisfactory understanding of the suitability of visualization methods for different types of data.	Attains a very good understanding of a range of visualization methods and their suitability for different data types.	Achieves an exceptional understanding of advanced visualization methods and their nuanced suitability for different data types.
Demonstrate a critical understanding of Data Science Ethics	Struggles to demonstrate a basic understanding of data science ethics.	Displays some understanding of data science ethics but falls short of a critical analysis.	Demonstrates a satisfactory understanding of data science ethics.	Attains a very good understanding of a range of ethical considerations in data science.	Achieves an exceptional understanding of advanced ethical considerations in data science.

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